Biodiversity

Genetically Modified-crops: advantages and disadvantages

Genetically modified (GM) crops:

GM crops are genetically improved and contain a gene or genes from the same or a different species artificially inserted in its genome. Genetically modified crops (GM crops) are **plants used in agriculture, the DNA of which has been modified using genetic engineering methods**. GM crops contain gene(s) artificially inserted instead of the plant acquiring it through pollination or other natural methods.

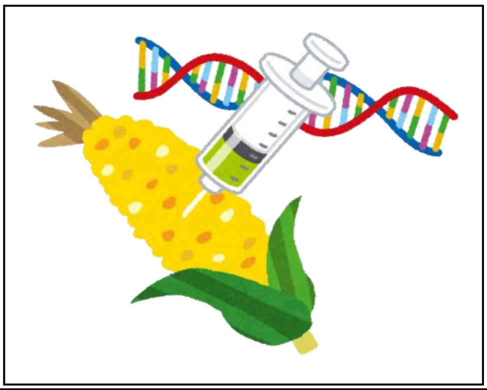

Fig: - Genetically modified crops

The first GM plant was introduced in 1982, which was an antibiotic-resistant tobacco plant. The first commercially produced GM plant was introduced in the US in 1994, the *FlavrSavr* tomato, which had longer shelf-lives.

Transgenic crops and conventionally-bred crops can directly affect the environment in different ways which include: gene transfer to wild relatives or conventional crops, weediness and trait effects on non-target species. Transgenic crops can also indirectly affect the environment as they have specific requirements in terms of pesticide and herbicide use and cropping patterns which requires changes to be introduced in existing agricultural practices.

Transgenic trees are a cause of concern for the environment, more so because of their long-life cycle. Transgenic micro-organisms used in food processing are normally used under confined conditions and are generally not considered as environmental risks. Some kinds of microorganisms can be used in the environment as biological control agents or for bioremediation of environmental damage (e.g., oil spills). The implications to the environment must be assessed before such organisms are released. The main concern with transgenic fish is their potential to breed with and out- compete wild relatives. Transgenic farm animals on the other hand, are generally reared in highly confined conditions and therefore do not pose a risk to the environment.

What is the difference between Traditional vs GM

Traditional breeding methods:

Traditional plant breeding techniques allows for gene exchange via transfer of male (pollen) of one plant to the female organ of another.

Disadvantages

- This method limited to exchanges between same or very closely related species.
- Time consuming to achieve desired set of traits, which may or may not be available in related species.

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Advantages of GM technology

GM technology enables plant breeders to bring together useful genes for the creation of superior plant varieties, from a wide range of living sources, not limited to closely related species.

Benefits of GM crops:

1. Improved nutritional value
2. Toxin reduction
3. Stress resistance
4. Useful by-products
5. Bioremediation

1. Improved nutritional value:

The nutritional content of the crops can be altered as well, providing a more nutritional profile than what previous generations were able to enjoy. This means people in the future could gain the same nutrition from eating lesser amounts of food. For example, rice can be genetically modified to produce high levels of Vitamin A. This can help reduce global vitamin deficiencies.

2. Toxin reduction

Potato that prevents bruising and produces lesser acrylamide on frying.

3. Stress resistance:

One of the main advantages of GM technology is that crops can be engineered to withstand weather extremes. This means that there will be good quality and sufficient yields even under poor or severe weather conditions. Herbicide resistance, pest resistance, resistance to cold. Plants capable of withstanding stressors like draught, frost, high soil salinity.

E.g., DroughGard maize: draught resistant maize, introduced in the US.

Fig: - DroughGard maize

4. Useful by-products

- Plants engineered to produce useful by-products such as drugs, biofuel (algae), bioplastics

5. Bioremediation:

Bioremediation is a process used to treat contaminated media, including water, soil and subsurface material, by altering environmental conditions to stimulate growth of microorganisms that degrade the target pollutants. GM plants for bioremediation of contaminated soils containing Hg, Se, PCBs, TNT, RDX etc.

e.g., switchgrass and bentgrass

Other advantages:

If we are using GM foods, there is several other advantages also there. Like.

- Cheaper and faster to grow crops.
- GMO crops are bred to grow efficiently. This means that farmers can produce the same amount of food using less land, less water, and fewer pesticides than conventional crops. Because they can save on resources, food producers can also charge lower prices for GMO foods.
- **Easier to transport:** Because GMOs have a prolonged shelf life, it is easier to transport them greater distances. GMO food gives us the opportunity to limit food waste, especially in the developing world, so that hunger can be reduced and potentially eliminated. No more malnutrition or lack of availability of food.
- **Endless possibilities:** anything alive can be genetically modified.

Gene Flow:

In the short term, the spread of transgenic herbicide resistance via gene flow can lead to logistical and/or economic problems for farmers. In the long run, transgenes that confer resistance to pests and environmental stress and/or lead to greater seed production are most likely to favour weeds or have a negative impact on non-target species. A number of transgenic traits have the potential to contribute to sustainability in agricultural systems. The benefits and risks associated with the use of transgenic crops need to be studied carefully in a comprehensive manner and systematically analysed. There is an urgent need to make this exercise a top priority.

Environmental risks of genetically modified organism (GMO):

1. Unexpected gene flow
2. Horizontal Gene Transfer
3. Competition with natural species
4. Increased selection pressure on target and non-target organisms
5. Ecosystem impacts

1. Unexpected gene flow:

Interbreeding between genetically modified organism (GMO) and wild type weeds and/or related species, can result in uncontrollable or **irreversible escape of genes** into neighboring wild plants by pollen.

E.g.: Hybrid rice crossbreeding with a weedy relative, confers on the latter, the competitive advantages of higher photosynthetic rates, more shoots, flowers and seeds.

2. Horizontal Gene Transfer:

The transfer of **foreign genes** to other organisms such as bacteria/virus that can cause harm to environment.

E.g.: Transfer of an antibiotic resistance gene to a pathogen can be terrible to humans/animals.

3. Competition with natural species:

Genetically modified organism (GMO) has favorable traits built-in, such as higher yields or resistance to environmental stress, presenting them with a natural advantage over native organisms, allowing them to become invasive, spread into new habitats unchecked and cause ecological damage.

4. Increased selection pressure on target and non-target organisms:

Evolution of resistant pests and weeds, termed **superbugs**, in response to herbicide-resistant crops. Constant spraying of herbicide on such crops would result in **acquired resistance** by surrounding weeds, resulting in a higher dose of the same, or a different type.

5. Ecosystem impacts:

- Genetic modification produces genetically modified animals, plants and organisms. If they are introduced into the environment, they can affect biodiversity. For example, existing species can be overrun by more dominant new species.
- Effect of a single species may extend beyond a single ecosystem, carrying with it risks of ecosystem damage and destruction.

In summary,

Advantages of GMO's:

- Enhance desired traits
- Pest resistance
- Improve nutritional content
- Less time than controlled breeding
- Improve accuracy
- Herbicide tolerance
- Cold tolerance

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- Medical advantage e.g., Edible vaccines
- Virtual end of world hunger. E.g., No malnutrition
- Cheaper or faster to grow and don't have to be rich in plant
- Endless possibilities and anything alive can be genetically modified
- Reduce production cost to reduced chemical and mechanical need in planting, maintenance and harvest.

Disadvantages of GMO's:

- Harm to organisms
- Does not taste natural
- Spread of superweeds
- Spread of superbugs
- New trade, tariff and quota issues
- May cause health problems ϖ Larger companies have more power
- Possible greed to GMO manufacturers
- Unforeseen allergen risks
- Allergies may become more intense
- New allergies may arise
- Widening corporate size gaps between food producing giants and smaller ones.

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